

Reported skill in daily river water-temperature prediction shrinks under strong baselines and whole-region holdout

[AUTHOR LIST TO BE COMPLETED]¹

¹[AFFILIATION 1 TO BE COMPLETED — department or laboratory, institution, city, postcode,
country]

²[AFFILIATION 2 TO BE COMPLETED — add or delete affiliation lines to match the final author list]

Key points

- Station-median skill at a seven-day lead is +0.251 against persistence but +0.038 against damped persistence at 120 U.S. gauges.
- A gradient-boosted tree with site identity has the lowest station-median RMSE at 1, 3, and 7 days on identical prediction keys.
- Holding out whole hydrologic regions instead of random sites lowers three-day skill against persistence from +0.187 to +0.116.

Corresponding author: [CORRESPONDING AUTHOR TO BE COMPLETED],
[INSTITUTIONAL E-MAIL TO BE COMPLETED]

Abstract

Daily river water temperature is strongly persistent and strongly seasonal, and machine-learning studies routinely report large skill gains for it. Those gains are almost always measured against naive persistence, climatology, or an air–water regression, on random data splits, with each model scored on its own set of predictable days. Such designs cannot separate learned river behaviour from three cheaper explanations: seasonal damping, favourable key selection, and spatial interpolation between neighbouring gauges. Here we score a constrained deep predictor against damped persistence, gradient-boosted trees with and without site identity, a global LSTM, and an information-matched plain causal convolutional network on one common set of 249,072 station/date/horizon keys, at 120 U.S. gauges in 15 hydrologic regions over 2006–2020, with all preprocessing fitted strictly backwards in time and whole regions, not random sites, held out. Median station skill (one minus the RMSE ratio) against persistence is +0.251 at a seven-day lead but +0.038 against damped persistence: of the 0.560 °C reduction in station-median RMSE from persistence to the deep model, 0.479 °C is delivered by damping alone. A gradient-boosted tree has the lowest station-median RMSE at all three leads, and the information-matched plain convolutional network is within 0.023 °C of the full architecture. Three-day skill against persistence falls from +0.187 under a random held-site split to +0.116 under whole-region holdout, at a mean nearest-training-gauge distance of 289 km. Reported skill for this variable is therefore largely a statement about the reference model and the spatial partition rather than about the architecture.

Plain Language Summary

River temperature changes slowly from day to day, so a forecast that repeats yesterday's reading is already fairly accurate, and one that also nudges it toward the usual value for the time of year is better still. That makes it easy to publish a model that looks impressive without being useful. We tested a river-temperature model at 120 U.S. gauges over fifteen years and deliberately made the test hard: every model was scored on exactly the same days and sites, none could see information from after the moment it was asked to predict, and we held out entire river regions rather than scattered gauges. Measured the usual way, our model looked strong. Measured against a simple seasonal adjustment of yesterday's reading, almost all of that advantage disappeared, and a standard

tree-based method was more accurate at every forecast range. How well such a model is reported to perform depends mostly on what it is compared against and on how the map is divided, not on how the model is built.

Keywords: river water temperature; benchmark design; strong baselines; spatial holdout; clustered inference; conformal prediction.

1 Introduction

Stream and river thermal regimes set dissolved-oxygen saturation, metabolic rates, life-stage timing, and habitat suitability, and they respond to atmospheric forcing, discharge, groundwater exchange, riparian shading, channel geometry, and regulation (Caissie, 2006). Water managers need daily-resolution temperature at gauged reaches to time releases, anticipate thermal refuge loss, and interpret compliance monitoring, and the supply of daily records from national monitoring networks has made the variable one of the most heavily modelled targets in applied hydrological machine learning.

The prevailing understanding in that literature is that deep sequence models have substantially advanced daily river-temperature prediction. Nonlinear air–water regressions (Mohseni et al., 1998) and hybrid air-temperature and discharge formulations (Toffolon and Piccolroaz, 2015; Piccolroaz et al., 2016) were succeeded by recurrent, multi-task, and physics-guided river-network architectures that report large error reductions (Feigl et al., 2021; Rahmani, Lawson, et al., 2021; Jia et al., 2021; Sadler et al., 2022; Zwart et al., 2023). The reported gains are consistent enough that architectural sophistication is now the usual explanation for accuracy in this problem.

Those reported gains are, however, extraordinarily heterogeneous, and the heterogeneity does not organise itself by architecture. Reviews of the field note that published accuracies are not comparable across studies because reference models, key sets, and data-splitting conventions differ (Corona and Hogue, 2025). The dispersion is larger than the architectural differences it is supposed to explain, and it survives within single panels: on the data used here, one fixed model and one fixed set of prediction days produce a seven-day skill of +0.251 or of +0.038 depending only on

which reference model is placed in the denominator. An architectural account cannot produce that spread, because the architecture does not change between the two numbers.

The identification gap is a property of the designs, not of the literature's size. Three design choices are made almost universally, and each one absorbs an alternative explanation into the reported number. First, skill is quoted against naive persistence, climatology, or an air–water regression. Daily mean water temperature has large thermal inertia, so relaxing the last observation toward a seasonal climatology already removes most of the error a learned model can remove; a reference that omits this hands the model a large part of its budget before it learns anything. Second, models are scored on model-specific complete-case sets, so a model can gain apparent accuracy by declining to predict on hard days, and preprocessing statistics — scaling constants, imputation fills, climatological references, event thresholds, interval offsets — are commonly fitted on periods that include the evaluated interval, which transfers information backwards in time without leaving a trace in a conventional split. Third, spatial generalisation is reported from random held-site splits. When a held-out gauge's neighbours remain in training, the model reaches the held-out regime through correlated forcing and spatially smooth representations; the quantity measured is closer to interpolation than to transfer. None of these choices is detectable from a reported RMSE, and each inflates it.

Here we hold the model and the data fixed and vary the three design choices instead, so that their separate contributions to reported skill become measurable. We assembled a frozen panel of 657,480 site-days from 120 stable U.S. Geological Survey site numbers spanning 2006–2020, 34 states, and 15 two-digit hydrologic unit (HUC2) regions, and evaluated one common set of station/date/horizon keys at 1-, 3-, and 7-day leads. Four questions organise the study. How much of a reported gain survives a strong reference? Which model class is actually most accurate on identical keys? How much of a reported spatial transfer survives whole-region holdout? And how far do those answers vary across regions, seasons, and the width of a calibrated interval? The predictor whose skill we decompose, ThermoRoute, is described in full in Section 3.1; its architecture is the object under test rather than the contribution, and the evaluation-period labels covering 2021–2023 are acquired exactly once, after the model suite and analysis code are frozen and replayed.

2 Data and design

2.1 The frozen panel and the cohort it defines

The development record is a single derived panel, `data_usgs/panel_usgs_120v2.parquet`, bound by a manifest (`data_usgs/frozen_panel_v1.json`) to a station registry (`data_usgs/station_registry_v1.csv`). Each of 120 sites carries one row for every calendar day from 2006-01-01 through 2020-12-31, giving 657,480 rows with no gaps in the date index; missing observations are explicit nulls rather than absent rows. Station identity is the zero-padded USGS `site_no`.

The seven raw variables consumed by the models, in frozen order, are water temperature (`WTEMP`), streamflow (`FLOW`), air temperature (`TEMP`), precipitation (`PRCP`), a relative-humidity proxy (`RHMEAN`), Daymet daylight-period mean incoming shortwave radiation in W m^{-2} (`DH`), and wind speed (`WDSP`). Two names are legacy and easily misread: `DH` is Daymet `srad`, the incident shortwave flux averaged over the daylight period, not day length and not a 24-hour mean; `RHMEAN` is a reproducible vapour-pressure proxy evaluated at the `tmax/tmin` midpoint, not a direct daily-mean humidity observation. Gage height (`WLEVEL`) is retained as raw evidence but is not a model input. Water temperature and discharge come from USGS NWIS daily values (U.S. Geological Survey, 2024), meteorological fields from Daymet V4 (Thornton et al., 2022), and wind from gridMET (Abatzoglou, 2013).

The cohort was not sampled at random from U.S. rivers, and nothing here treats it as though it were. Candidate selection began from a discovery query over stream sites in the states represented by the panel and rejected 1,345 of 1,465 candidate stations: 950 lacked joint water-temperature and discharge availability, 376 failed a full-period coverage threshold, and 19 failed a 2019–2020 coverage threshold. The retained cohort is therefore availability-enriched, and the 2019–2020 interval participated in its construction, so that interval is development data in the strict sense and is exploratory throughout this manuscript.

Three properties of the retained cohort bear directly on the analysis. Observed water temperature is absent on about 15.8% of panel rows, with a maximum contiguous gap of 2,404 days, and discharge on about 2.8%, with a maximum gap of

1,369 days; the meteorological fields are missing on about 0.07% of rows, in a structured rather than random way, because all 120 sites lack the provider calendar's final day in the leap years 2008, 2012, 2016, and 2020. Among retained stations, 38 share a repeated hydrologic unit code with another retained station and 19 have another retained station within 10 km, so station-level independence is not tenable. Two stations contain 2,059 rows of signed negative discharge (minimum -121 cfs), retained with their sign rather than clipped.

2.2 Temporal roles and the common key set

Temporal roles are fixed before any model is fitted, and a training sample is admitted only if its issue date and every one of its target dates fall inside the same partition.

Role	Dates	Rows	Observed WTEMP	Permitted use
Training	2006–2015	438,240	341,646	fit preprocessing and models
Validation	2016–2017	87,720	84,074	choose frozen settings
Calibration	2018	43,800	42,279	conformal and Platt fits; final year of the 2006–2018 seasonal event reference
Development evaluation	2019–2020	87,720	85,621	previously inspected; exploratory diagnosis only
One-time evaluation	2021–2023	—	sealed	opened once, after all freezes (Section 3.7)

On the development-evaluation partition, the intersection of admissible keys across all primary models comprises 249,072 station/date/horizon keys, evenly distributed as 83,024 keys per lead. Every development comparison in Section 4 uses exactly this key set. No model-specific complete-case set is permitted, so a model cannot gain apparent accuracy by declining to predict on hard keys.

2.3 Evaluation-period inputs and the information boundary

The evaluation interval is 2021-01-01 through 2023-12-31 for the same 120-site cohort, and historical Daymet and gridMET covariates for that interval are retrieved and archived before any outcome is accessed. Meteorology is represented at each station coordinate and is not aggregated over upstream catchments, which is a real limitation for large basins (Section 6.2).

The primary information set uses provider values dated no later than each historical issue date and consumes no horizon-specific future weather forecast. This is a date-indexed retrospective hindcast, and not a re-execution of what a forecaster could have run on the day (Section 6.1): the as-issued provisional vintage of a gridded product cannot be reconstructed after the fact, so archiving requests, responses, timestamps, and checksums freezes the dataset actually evaluated without proving that identical values were available operationally at the time. NWIS dates denote site-local finalized daily values while Daymet and gridMET use provider-specific calendar-day definitions; no subdaily day-boundary harmonization is claimed. A separate outcome-free bridge, described in the Supporting Information, re-fetched 2018–2020 covariates with the parser used for the evaluation-period acquisition and records `PASS_EXACT_PRODUCT_BRIDGE` on the exact site/date registry.

A restricted external cohort is defined after the model suite is committed: a fixed 34-state metadata query identifies stream sites advertising daily water-temperature capability, without requesting values, coverage dates, or event rates, and a deterministic seed selects 30 stations whose site identifiers do not occur in the development registry. This is site-identifier disjointness only, and those models still consume each target site's observed history (Section 3.4).

3 Methods

3.1 The predictor

ThermoRoute predicts at issue time t , station i , and lead h by adding a bounded learned correction to a damped-persistence anchor. One model is fitted for all three leads jointly: the horizon set (1, 3, 7 days) is an input dimension and

every head emits a lead-indexed vector in a single forward pass, so no lead has its own weights.

Anchor. A frozen seasonal climatology $\mathbf{c}$ and a per-station decay $\varphi_{\mathbf{i}}$ give

$$(1) \mathbf{A}_{\{\mathbf{i}, \mathbf{t}+\mathbf{h}\}} = \mathbf{c}_{\{\mathbf{i}, \mathbf{t}+\mathbf{h}\}} + \varphi_{\mathbf{i}}^{\mathbf{h}} \cdot (\mathbf{y}_{\{\mathbf{i}, \mathbf{t}\}} - \mathbf{c}_{\{\mathbf{i}, \mathbf{t}\}})$$

where $\mathbf{y}_{\{\mathbf{i}, \mathbf{t}\}}$ is the last observed daily mean water temperature. Each $\varphi_{\mathbf{i}}$ is estimated by no-intercept least squares on consecutive-day anomaly pairs from the training period only, requires at least 30 pairs, and is clipped to $[0, 0.999]$. This object, fitted identically, is also the damped-persistence reference.

Learned relaxation proposal. A station-, flow-, and season-conditioned relaxation rate replaces the fixed decay in a separate proposal path:

$$(2) \kappa_{\{\mathbf{i}, \mathbf{t}\}} = \sigma(\mathbf{b} + \mathbf{b}_{\mathbf{i}} + \beta_{\mathbf{q}} \mathbf{q}_{\{\mathbf{i}, \mathbf{t}\}} + \beta_{\mathbf{s}} \mathbf{s}_{\mathbf{t}}), \text{ clipped to } [10^{-3}, 0.999]$$

$$(3) \mathbf{P}_{\{\mathbf{i}, \mathbf{t}+\mathbf{h}\}} = \mathbf{c}_{\{\mathbf{i}, \mathbf{t}+\mathbf{h}\}} + \mathbf{e}_{\{\mathbf{i}, \mathbf{t}\}} + (1 - \kappa_{\{\mathbf{i}, \mathbf{t}\}})^{\mathbf{h}} \cdot (\mathbf{a}_{\{\mathbf{i}, \mathbf{t}\}} - \mathbf{e}_{\{\mathbf{i}, \mathbf{t}\}})$$

with $\mathbf{a}_{\{\mathbf{i}, \mathbf{t}\}} = \mathbf{y}_{\{\mathbf{i}, \mathbf{t}\}} - \mathbf{c}_{\{\mathbf{i}, \mathbf{t}\}}$ the current anomaly, $\mathbf{e}_{\{\mathbf{i}, \mathbf{t}\}}$ a learned equilibrium anomaly formed from the standardized forcings and a station term, $\mathbf{q}$ the standardized log-discharge, and $\mathbf{s}_{\mathbf{t}}$ a two-component season encoding. The bias $\mathbf{b}$ is initialised at -2.94 , so the proposal begins near damped persistence. The fitted relaxation rate is a statistical quantity only: it is not a heat-transfer coefficient, and no physical interpretation of its value is offered anywhere in this paper.

Router. A sparse, horizon-conditioned router scores all 7 variables at lags 0 to 14 — 105 (variable, lag) pairs — with a scaled dot product between a horizon-specific query and a per-pair key, normalised by sparsemax (Martins and Astudillo, 2016):

$$(4) \mathbf{w}_{\mathbf{h}} = \text{sparsemax}(\langle \mathbf{q}_{\mathbf{h}} + \mathbf{ctx}, \mathbf{k}_{\{\mathbf{v}, \ell\}} \rangle / \sqrt{\mathbf{d}}), \text{ over } \mathbf{v} \in 1 \dots 7, \ell \in 0 \dots 14$$

The router is an allocation mechanism inside the predictor, not a map of connected reaches; it receives no verified graph or topology input.

Temporal encoder. A strictly left-looking temporal convolutional network (Bai et al., 2018) of $B = 2$ residual blocks with kernel $k = 3$, dilations 1 and 2, and width 40 encodes recent history. Left padding of $(k - 1) \cdot 2^b$ makes the output at position t a function of positions $\leq t$ by construction. The receptive field is

$$(5) R = 1 + (k - 1)(2^B - 1) = 7 \text{ steps}$$

so, together with the router's oldest usable value at lag 14, no input older than lag 14 can affect the output. The sequence builder supplies a 32-day tensor, but that is a construction buffer, not an effective-memory claim. Throughout this paper "causal" describes time ordering only and never causal inference.

Regime mixture. Three experts are combined by a soft gate (Shazeer et al., 2017):

$$(6) r = \sum_{k=1}^3 \pi_k \cdot E_k(\cdot), \pi = \text{softmax}(g(\cdot))$$

Bounded residual. The learned point correction is squashed and rescaled, so its deviation from the anchor is algebraically bounded:

$$(7) \hat{y}_{\{i,t+h\}} = A_{\{i,t+h\}} + \delta \cdot \tanh(z_{\{i,t+h\}} / \delta), \delta = 1.0 \text{ } ^\circ\text{C}$$

giving $|\hat{y} - A| < \delta$ identically. The bound is relative to the named anchor: it limits the point output's deviation from damped persistence and bounds neither absolute error, nor event-tail error, nor interval width, nor behaviour after a distribution shift. Because the 2019–2020 partition was inspected during development, $\delta = 1.0 \text{ } ^\circ\text{C}$ is a development-selected algebraic point bound and is not a selection from an untouched holdout.

Loss. With y the observed target, A the anchor, quantile levels $\tau \in \{0.05, 0.50, 0.95\}$, Q_τ the corresponding quantile head, and ρ_τ the pinball loss,

$$(8) L = \text{MSE}(y, \hat{y}) + \sum_\tau \rho_\tau(y, Q_\tau) + \lambda_e \cdot \text{BCE}(\text{exceedance}) + \lambda_c \cdot C + \lambda_r \cdot \|\hat{y} - A\|_1$$

with $\lambda_e = 0.3$, $\lambda_r = 10^{-2}$, $\lambda_c = 1.0$, and a temperature loss scale of 1.0. The point and pinball terms carry unit weight. The term C penalises quantile non-

monotonicity and is identically zero under the head parameterisation used here, so it contributes no gradient; it is retained because it was declared before fitting.

Fitting. All neural members are fitted with AdamW (Kingma and Ba, 2015; Loshchilov and Hutter, 2019) at learning rate 2×10^{-3} , weight decay 10^{-4} , batch size 1,536, gradient-norm clipping at 1.0, a maximum of 80 epochs with early stopping at patience 12 on station-macro validation RMSE, a plateau scheduler halving the rate at patience 4, and equal-station fixed-size bootstrap sampling; five seeds (0–4) are fitted and averaged with equal weights. The canonical configuration has **38,505 trainable parameters**. All members were fitted on CPU only (Intel Xeon Gold 6430, one PyTorch intra-op and one inter-op thread), and every seed stopped early, at epochs 18, 15, 14, 18, and 26. Measured end-to-end training wall-clock time is [TRAINING WALL-CLOCK TO BE RECORDED] and inference throughput is [INFERENCE COST TO BE RECORDED]; neither is present in the run records, and neither is estimated here.

3.2 The reference set

The composition of the reference set is the most consequential methodological choice in this study, because it is what the reported gain is a gain over. Each member removes one alternative explanation for apparent skill.

Persistence ($\hat{y}_{\{t+h\}} = y_t$) quantifies the raw memory of the series.

Damped persistence, equation (1) fitted on the training period, absorbs the two effects a learned model most easily rediscovers — inertia and seasonality — and is the reference against which the primary comparisons are stated. **Seasonal climatology** isolates the seasonal cycle alone. **LightGBM** (Ke et al., 2017) is the principal learned reference and is run both as a global model receiving stable site identity as a categorical feature and as a per-station variant that isolates the value of pooling. A **global LSTM** with a station embedding represents the deep sequence family that has produced the strongest recent results for this variable (Rahmani, Lawson, et al., 2021; Zwart et al., 2023). A **plain causal temporal convolutional network** is the information-matched deep reference: it receives the same outcome-free issue-time history, station identity, climatology, frozen damped anchor, standardized environmental context, and calendar and regime-gate tensors as the full

model, is matched to within 2% of its parameter count (38,346 against 38,505) and to the same optimiser-step budget, and predicts an unrestricted residual around the same anchor while omitting the relaxation proposal, router, mixture, and residual bound. Its companion, a plain multilayer perceptron at 38,860 parameters, isolates the value of sequence structure. Neither ever reads the target tensor, the target-date field, or the disabled water-level channel.

The learned references are given a genuine chance to win. The global LSTM uses the same development splits, loss, station-balanced sampling, epoch and patience limits, history length, site identity, and five-seed budget as ThermoRoute, its predeclared validation-only grid may expose the same frozen climatology, damped-anchor, and season features, and it is eligible for the same calibration; it lacks only the bounded-residual constraint and the lag router. LightGBM selects among four predeclared settings separately by lead on the 2016–2017 partition using station-macro RMSE alone; the LSTM selects among three predeclared architectures with seed 0 before fitting five members. Tuning budgets are documented but *not* equalized across model classes, which is a real asymmetry: any statement about LightGBM here is scoped to this frozen four-candidate procedure and this feature schema, not to gradient boosting in general. Seven one-factor architecture controls (DampedPriorOnly, TR-noDynamicPrior, TR-fixedKappa, TR-noRouter, TR-noMoE, TR-noTCN, TR-unbounded) are fitted with the same five seeds and paired within seed on identical keys before ensemble averaging; they are deletion and intervention sensitivities and do not prove component necessity or identify a mechanism.

An air2stream-style hybrid reference was declared but is recorded as NOT_RUN, and the available implementation is an unofficial style reference rather than the published code or a validated reproduction of it. It is therefore excluded from every claim in this paper; its status, provenance, and what a defensible comparison would require are reported in the Supporting Information rather than carried as an unfilled column.

3.3 Leakage control

Leakage control here is a set of mechanisms that can be checked against the code and the artifacts, rather than a statement of intent.

Issue-time separation of covariates. Every predictor consumed by a primary model carries a date no later than the issue date, and no horizon-specific future weather field enters any primary model.

Non-anticipating sequence encoding. Equation (5) makes the information horizon bounded and auditable by construction rather than by convention.

Strictly backward-fitted statistics. Standardization constants, imputation fills, the seasonal climatology, the damped-persistence rate, the station-level q90 event thresholds (2006–2015), the seasonal event reference (2006–2018), the conformal offsets (2018), and the Platt calibrators (2018) are each fitted on data strictly preceding the interval on which they are applied. This closes the most common backward-information channel in hydrological benchmarking, which is not the model but the preprocessing.

Explicit admissibility. A forecast key is admissible only when issue-date water temperature is genuinely observed and a 32-day history can be constructed; other history cells are filled with training-only seasonal medians and retain explicit missingness masks. We record one honest gap: the issue-date auxiliary quantities used by the relaxation proposal and the regime gate are computed from the same training-only imputed panel but do not each carry a separate auxiliary-path mask, so the output is not guaranteed invariant to the chosen fill values.

Replay under isolation. Before any label acquisition is authorized, a fresh isolated interpreter, denied network access, child processes, repository writes, and reads from the evaluation namespaces, reloads every trained member and reproduces the validation, calibration, and 2019–2020 keys and values from the frozen inputs, independently recomputing the thresholds and refitting the calibration parameters; a prediction file that is merely self-consistent is rejected. The isolation model, the byte bindings, and the publication and crash semantics of that replay are specified in the Supporting Information.

3.4 Two spatial partitions

Random held-site splits are the usual way to report spatial generalisation in this literature, and they leak. If a held-out gauge's neighbours remain in training,

the model reaches the held-out site's thermal regime through correlated forcing, shared preprocessing statistics, and spatially smooth learned representations. We therefore report two arms with different leakage properties and read the difference between them as the measurement of interest.

The **random held-site warm-start** arm uses four folds in which held-site histories still contribute to global panel preprocessing. This is the arm most comparable to published random-split results, and precisely for that reason it is not the arm we treat as the spatial test.

The **held-region gauged transfer** arm packs the 15 HUC2 groups into four folds of [30, 30, 31, 29] stations and holds out *whole regions*, so no gauge from a held-out region appears in training. Station-agnostic ThermoRoute and a global LightGBM are each trained on the in-fold regions and forecast the held-out region's stations, with all climatology, scaling, and damped-rate parameters pooled from in-fold stations only. The mean distance from a held-out station to its nearest training gauge is 289 km.

The label on this arm must be exact. Held-out sites still supply their own observed water temperature through the issue date, both as the persistence anchor and as the sequence history. This is **gauged** transfer to a region whose gauges were not used in fitting; it is **not** prediction at a site with no observational record, which is a different problem and a different evaluation (Weierbach et al., 2022; Rahmani, Shen, et al., 2021). The same caveat applies to the 30-site external cohort of Section 2.3.

3.5 Conformal intervals

Predictive uncertainty in hydrological modelling has most often been represented through Bayesian or likelihood-based treatments of parameter and input error (Beven and Binley, 1992; Kavetski et al., 2006a, 2006b). We use a distribution-free calibration step instead, because the quantity required here is an empirical coverage statement about a fixed set of forecast keys rather than a posterior over model parameters. Intervals come from split conformalized quantile regression (Romano et al., 2019; Vovk et al., 2005): the learned models emit a separate mean-squared-error point head and three pinball-trained quantile heads, members are averaged with equal weights, and the offset is fitted on the 2018 calibration year only. The con-

struction is deliberately one-sided — the deployed offset is `qhat_plus = max(raw_qhat, 0)` and the delivered interval is `[q05 - qhat_plus, q95 + qhat_plus]` with the median unchanged — so calibration can widen a nominal interval but never shrink it, and a non-finite, crossed, or empty final interval fails closed before any evaluation label is read.

What this guarantees is worth stating precisely, because conformal prediction is frequently over-claimed in applied work. Split conformal provides finite-sample *marginal* coverage under exchangeability of calibration and evaluation nonconformity scores. Here the calibration scores come from 2018 and the evaluation scores from a later, disjoint period at the same sites, so exchangeability fails by construction: there is temporal drift, and the scores are dependent within station and within region. We therefore report achieved coverage as an *empirical marginal diagnostic*, make no finite-sample guarantee, and make no conditional-coverage claim of any kind. Two pre-specified sensitivities probe that boundary: a block-maximum variant that calibrates on the maximum score within groups of consecutive retained calibration rows, following the standard practice of resampling blocks rather than rows when the series is dependent (Künsch, 1989), and an idealized delayed adaptive-conformal variant using each forecast's target date as a feedback-arrival proxy. Neither replays real feedback availability, because the inputs contain no verified observation-publication timestamp, revision history, or reporting latency. The equal-weight three-quantile pinball summary is not CRPS; the two are distinct members of the family of proper scoring rules (Gneiting and Raftery, 2007). An event head separately reports exceedance of each station's 2006–2015 q90 water temperature, calibrated by one Platt map per lead fitted on 2018 only; that threshold is a statistical tail diagnostic local to each station, with no biological, ecological, or regulatory interpretation.

3.6 Estimand, metrics, comparison set, and why this study is descriptive

Two differently signed quantities are reported, and they are never combined. The paired effect is

(9) $\Delta\text{RMSE} = \text{RMSE}(\text{candidate}) - \text{RMSE}(\text{reference})$, in $^{\circ}\text{C}$, **negative**

favours the candidate

and the skill score is

(10) $\text{skill} = 1 - \text{RMSE}(\text{candidate})/\text{RMSE}(\text{reference})$, dimensionless,

positive favours the candidate

Equation (9) is the formal estimand and carries a $^{\circ}\text{C}$ unit everywhere it appears; equation (10) is a ratio and appears as a bare signed number. Every value in Section 4 is labelled with which of the two it is. A positive ΔRMSE and a positive skill score mean opposite things, and no figure axis, table column, or sentence in this paper mixes them.

The sampling unit is the station. For each lead, unweighted RMSE is computed on the common daily keys separately for each reportable station, and the primary effect is the unweighted median across stations of equation (9) with ThermoRoute as candidate; a station/lead cell is reportable only with at least 100 valid paired targets, so daily rows do not determine between-station weight. We report RMSE and paired RMSE differences rather than an efficiency-type criterion such as the Nash–Sutcliffe efficiency or its decomposition-based successors (Nash and Sutcliffe, 1970; Gupta et al., 2009), because those criteria normalise by a station-specific variance, which would make a paired between-model difference on identical keys harder to read.

The frozen comparison set contains exactly five rows:

#	Comparison	Lead	Predeclared margin
1	ThermoRoute vs. damped persistence	1 d	0.00 $^{\circ}\text{C}$
2	ThermoRoute vs. damped persistence	3 d	0.00 $^{\circ}\text{C}$
3	ThermoRoute vs. damped persistence	7 d	0.00 $^{\circ}\text{C}$
4	ThermoRoute vs. LightGBM	3 d	+0.05 $^{\circ}\text{C}$ numerical ceiling
5	ThermoRoute vs. LightGBM	7 d	+0.05 $^{\circ}\text{C}$ numerical ceiling

The +0.05 $^{\circ}\text{C}$ ceiling was fixed before outcomes as a strict numerical limit on allowable degradation. It is not derived from sensor precision, biological response, a

water-quality standard, or an elicited stakeholder utility, and it carries no ecological or regulatory importance.

Two uncertainty procedures accompany each row, both clustered at HUC2: a one-sided p-value from exact enumeration of all 2^K whole-cluster sign vectors, applying one common sign to every station effect within a cluster, and a 10,000-draw whole-HUC2 cluster bootstrap percentile interval for the median station effect, with Holm adjustment (Holm, 1979) over exactly these five p-values. We do not use the standard tests of equal predictive accuracy for forecast series (Diebold and Mariano, 1995; Harvey et al., 1997), because their asymptotics are stated for a single loss-differential series and do not address dependence across the 120 gauges, which is the dominant dependence here.

Those procedures cannot carry inferential weight in this study, for three independent and permanent reasons. Eligibility is a fixed property of the frozen gate rather than a quantity recomputed from the cohort, and the opening refuses to run unless the gate has failed closed. The reasons follow in order of how binding they are.

First, the two structural assumptions on which both procedures rest are recorded as unmet. Whole-cluster resampling assumes independent, exchangeable sampling of HUC2 regions, and whole-cluster sign flipping assumes joint sign symmetry of each complete cluster effect vector around the tested margin. The frozen gate records both as permanently unmet, for reasons that are design facts rather than measurements: the regions were not probability-sampled, and no randomized sign assignment exists anywhere in the design. This is not a threshold a larger or differently partitioned cohort would clear. It is a statement about how the cohort came into being — an availability filter over gauges in 34 states — and it would hold with 30 regions as surely as with 15.

Second, the null-simulation component was never implemented. The gate requires a simulated-null calibration of the clustered procedures before they may be read inferentially. That component does not exist in the analysis code; it returns a fixed failure on every run and appends its own blocking reason. It is a declared requirement that was not built, disclosed rather than quietly dropped.

Third, and least binding, the cohort has too few clusters. The gate requires at least 30 reportable clusters, an effective-cluster fraction of at least 0.75, and a largest-cluster share below 0.25. Before any reportability attrition the 15 HUC2 groups contain between 2 and 26 stations, the largest holds 21.7% of stations, and the inverse-Herfindahl effective cluster count is 9.54, an effective fraction of about 0.636; two of the three components fail. We state plainly what this means: **the threshold is unreachable under a HUC2 partition for any U.S. cohort whatsoever**, because the Watershed Boundary Dataset defines about 21 HUC2 regions in total and the gate asks for at least 30. That is a specification error, made when the amendment was written and identified before any evaluation label was accessed — which is the only reason it can be reported here rather than discovered by a reader.

Disclosing the error is a better response than defending the threshold, and a far better response than subdividing the map until it passes. Subdividing *would* pass: recomputed with the gate's own code on the frozen registry, HUC2 gives 15 clusters at an effective fraction of 0.636, HUC4 gives 64 at 0.507, HUC6 gives 75 at 0.485, and HUC8 gives 95 at 0.758, clearing all three thresholds. We did not adopt HUC8. Adjacent HUC8 units on the same river are plainly not independent, so passing that way would satisfy the arithmetic while defeating its purpose; and it would change nothing, because eligibility is fixed by the first two reasons, so a HUC8 partition would clear the cluster gate and return the same verdict. Choosing the cluster definition after seeing which definition helps is precisely the failure mode the gate exists to prevent.

Because the frozen Route-A cohort contains at most 15 HUC2 groups (inverse-Herfindahl effective cluster count ≈ 9.54) and therefore fails the outcome-free inference gate requiring $n_clusters \geq 30$, $effective_cluster_fraction \geq 0.75$, and $largest_cluster_share < 0.25$, the formal five confirmatory comparisons are permanently restricted to fixed-cohort descriptive effects under `DESCRIPTIVE_ONLY_INFERENCE_GATE_FAILED` for confirmatory claim eligibility; exploratory and secondary analyses remain reportable as such and are not erased by the gate. Whole-HUC2 bootstrap intervals, exact sign-flip p-values, and Holm adjustments are assumption-conditional sensitivities only and must not be written as

confirmatory decision evidence; after opening they must not be rewritten as superiority, non-inferiority, equivalence, parity, or national / U.S.-river generalization.

Every one of the five rows is still rendered exactly once, including adverse, conflicting, or non-reportable outcomes, with the effect estimate, the win rate, the station and cluster counts, and the interval all shown. Secondary and exploratory analyses — a qualifier-restricted target sensitivity, equal-region and leave-one-region influence summaries, station-balanced probability diagnostics, architecture controls, and synthetic perturbations — are labelled as such and cannot promote or replace a formal row. Relative economic value in the cost–loss sense (Richardson, 2000) is recorded as `REV_NOT_EVALUATED_NO_PREDECLARED_COST_LOSS_RATIOS`, because no cost–loss ratios, management costs, actions, or stakeholder utility model were frozen. A predeclared coverage and calendar-balance audit accompanies each formal row with eight fixed descriptive candidates — equal weighting of the 12 year-by-season cells, leave-one-year for each evaluation year, and leave-one-season for DJF, MAM, JJA, and SON — of which the most adverse candidate-minus-reference effect is taken as the deterministic unfavorable value, with ties broken by frozen candidate order. That audit is restricted to keys where both issue-date and target-date water temperature are retained finite observations; it does not impute unavailable outcomes, test a missing-at-random assumption, or estimate all-calendar-day performance, and a favourable sensitivity can never rescue an unfavourable primary row.

3.7 The one-time 2021–2023 evaluation

The evaluation labels are acquired exactly once, and everything capable of being tuned is frozen and independently verified before that acquisition is permitted. Those labels may not be read by model-selection, feature-selection, threshold-selection, calibration, or station-inclusion code, and enforcement is ordered so that the order is itself a gate. The protocol and its seal, the claim renderer and validator, the source tree, the dependency locks, and all model members are frozen and exactly replayed under the isolation of Section 3.3; the model-suite registry is then committed *while* the external candidate registry and the evaluation-period predictor artifacts are still absent from the repository; and only then may metadata-only candidate discovery and retrospective predictor acquisition run. Authorization re-

plays the resulting version-control ancestry and content hashes to verify that the model-suite commit genuinely precedes them. Any source change after the model freeze invalidates authorization, and failure of the chronology gate does not produce a weaker result: it demotes the exercise to retrospective exploration.

The acquisition itself is narrow and irreversible. A raw-only child process using the USGS `dataretrieval` client (Hodson and Hariharan, 2023) is restricted to daily mean water temperature (00010, °C), discharge (00060, cfs), and raw-only gage height (00065, ft), and records exact request and response bytes, qualifiers, timestamps, and content hashes. Where two or more finite series exist for a parameter and date, the cell is marked missing with an explicit conflict code rather than averaged or selected, and **no site may be replaced after outcomes are accessed**. The primary analysis uses every finite parsed value regardless of approval qualifier, so qualifiers can never select or remove a station, model, date, or forecast key; a separately labelled sensitivity, fixed before opening, retains target keys only when the target qualifier has the exact token set $\{\mathbf{A}\}$. The acquisition is transactional, with honest-owner crash and replay guards rather than protection against a malicious owner or a same-privilege adversary; those details are in the Supporting Information.

The analysis protocol was written and frozen before any post-2020 outcome was requested or inspected, and fixes the cohort hashes, the temporal roles, the model registry, the estimand, the five-row comparison set and its margins, the clustered procedures, the multiplicity correction, the probabilistic and event contracts, the outcome quality-control policy, and the requirement that every row be reported exactly once regardless of direction; two later amendments and a probability-metric erratum are themselves outcome-free and separately sealed. We state the limit of that evidence plainly: the seals are repository-internal and assume an honest owner, with no external timestamp, public registration service, or independent custodian.

4 Results

Sections 4.1–4.5 report development-period quantities. **All of them are exploratory.** The 2019–2020 partition participated in cohort construction and has informed model and narrative development; it is not an independent test, and no

number in these subsections is decision-eligible. Unless stated otherwise, all development values are five-seed ensemble means on the 249,072 common keys, with station-median RMSE in $^{\circ}\text{C}$. Section 4.6 contains the evaluation-period slots.

4.1 The reference model, not the architecture, sets the reported gain

How much of a reported gain survives a strong reference? Most of it does not. On identical keys, the same fixed model reports a seven-day median station skill of +0.251 against persistence and +0.038 against damped persistence — a factor of 6.6 between two numbers that differ only in the denominator of equation (10).

Lead	Persistence	Damped persistence	LightGBM	LSTM	ThermoRoute
1 d	0.803	0.774	0.578	0.662	0.631
3 d	1.576	1.406	1.280	1.323	1.291
7 d	2.217	1.738	1.649	1.679	1.657

Station-median RMSE, $^{\circ}\text{C}$. Read down the seven-day row: station-median RMSE falls from 2.217 $^{\circ}\text{C}$ under persistence to 1.738 $^{\circ}\text{C}$ under damped persistence to 1.657 $^{\circ}\text{C}$ under ThermoRoute. Of the 0.560 $^{\circ}\text{C}$ total reduction, 0.479 $^{\circ}\text{C}$ is delivered by relaxing yesterday's observation toward a fitted seasonal climatology, and 0.081 $^{\circ}\text{C}$ by everything the deep model adds. The same arithmetic in skill units gives +0.203, +0.187, and +0.251 against persistence and +0.168, +0.076, and +0.038 against damped persistence at 1, 3, and 7 days, all dimensionless.

The gap widens with lead, which is the diagnostic signature of damping rather than of learned river behaviour: the longer the lead, the more of the error a seasonal relaxation removes on its own, and the less remains for a learned model to remove. A study reporting only the persistence column would describe this model as gaining a quarter of the seven-day error budget. A study reporting both columns describes the same fitted weights as gaining four percent.

Reported against the strong reference, the paired effects of equation (9) are -0.130 , -0.104 , and -0.062 $^{\circ}\text{C}$ at 1, 3, and 7 days, with ThermoRoute favoured at 0.86, 0.90, and 0.93 of the 120 stations, and exploratory whole-HUC2 cluster-

bootstrap intervals of $[-0.177, -0.081]$, $[-0.126, -0.083]$, and $[-0.071, -0.052]$ °C. Those intervals carry the 15-cluster structure discussed in Section 3.6, and every row of the development mirror is flagged `NO_STRONG_INFERENCE` with the warnings `SMALL_CLUSTER_COUNT_LT_30` and `LOW_EFFECTIVE_CLUSTER_FRACTION`. If the reference set determines the size of the reported gain, the next question is whether it also determines which model wins.

4.2 The most accurate model on this panel is a gradient-boosted tree

It does. The tree ensemble has the lowest development-period station-median RMSE at all three leads: 0.578, 1.280, and 1.649 °C against ThermoRoute's 0.631, 1.291, and 1.657 °C. The paired station-level differences (`ThermoRoute` – `LightGBM`, equation 9) are +0.046 °C at 1 day, +0.008 °C at 3 days, and –0.005 °C at 7 days, with ThermoRoute win rates of 0.00, 0.40, and 0.60 across the 120 stations. The one-day win rate is 0.00: not one of the 120 stations favours the constrained architecture at the shortest lead.

We report this as the primary result it is. On this panel, on identical keys, a well-tuned gradient-boosted tree with site identity is more accurate than the constrained deep architecture, decisively at 1 day and negligibly at 3 and 7, where the exploratory cluster-bootstrap intervals for the paired difference are $[+0.001, +0.020]$ and $[-0.010, +0.010]$ °C. The global LSTM is intermediate at 0.662, 1.323, and 1.679 °C. The architecture's remaining practical argument is not accuracy but the algebraic bound of equation (7), which held on 100.00% of audited rows, with a maximum absolute correction of 1.0000 °C against the configured 1 °C limit and the derived station-by-lead RMSE inequality holding in all 360 cells — a property of the construction rather than a fitted outcome.

4.3 An information-matched plain convolutional network reproduces the architecture

If a tree is more accurate, is any of the architecture's structure doing work? Very little of it is. Against the information-matched plain causal temporal convolutional network of Section 3.2 — same inputs, same anchor, same optimiser budget,

38,346 against 38,505 parameters — the full architecture's paired median station effect, computed within seed on identical keys, ranges across the five seeds from -0.0003 to -0.0105 °C at 1 day, -0.0077 to -0.0179 °C at 3 days, and -0.0035 to -0.0226 °C at 7 days. Every seed favours the full model; no seed by more than 0.023 °C. Against the plain multilayer perceptron the same effects range from -0.0360 to -0.0411 °C at 1 day, so sequence structure matters and the components layered on top of it do not.

The one-factor deletions agree. Removing the temporal encoder is the largest single effect, moving 1-day station-median RMSE from 0.631 to 0.679 °C and 7-day from 1.657 to 1.675 °C. Removing the router (0.634), the mixture (0.634), the dynamic prior (0.628), the fixed-relaxation constraint (0.627), or the residual bound (0.630) moves the 1-day figure by at most 0.004 °C in either direction. Replacing everything by the damped anchor alone gives 0.774 , 1.406 , and 1.738 °C. These are five-seed deletion and intervention sensitivities on paired keys; they do not prove component necessity or a capacity-matched attribution. Read together with Section 4.2, they locate this model's accuracy in the causal sequence encoder and in the anchor it corrects, not in the routing, mixture, or bounding machinery.

Comparability note. The controls in this subsection are per-seed values from the development-controls run and are compared only with each other and with the matched full-architecture arm from the same run (0.6452 , 1.3047 , 1.6682 °C). They are not the five-seed ensemble means of Sections 4.1–4.2 and must not be differenced against them.

4.4 Whole-region holdout removes a third of the transfer skill a random split reports

How much of a reported spatial transfer survives when the map is partitioned honestly? Between a quarter and a third of it does not.

Arm	Design	1 d	3 d	7 d
Temporal development	seen stations, 2019–2020	+0.203	+0.187	+0.251
Random held-site warm start	4 folds; held-site history in preprocessing	+0.178	+0.172	+0.241
Held-region gauged transfer	leave-HUC2-region-out	+0.155	+0.116	+0.155

Median station skill versus persistence, dimensionless. Against damped persistence the same three arms give +0.168 / +0.076 / +0.038 (temporal), +0.145 / +0.061 / +0.030 (random held-site), and +0.147 / +0.086 / +0.075 (held-region). The three-day skill against persistence falls from +0.187 to +0.172 when sites are held out at random and to +0.116 when whole regions are, and the seven-day figure falls from +0.251 to +0.241 to +0.155. The gap between the second and third rows is the direct measure of how much a random spatial split flatters a model on this panel: the random split retains 92% of the temporal-arm three-day skill, the regional split 62%.

The mechanism is neighbourhood, and the geometry supports that reading. Under whole-region holdout the mean distance from a held-out station to its nearest training gauge is 289 km, whereas in the intact cohort 19 of 120 stations have another retained station within 10 km and 38 share a hydrologic unit code with another retained station. A random held-site split on such a panel measures something much closer to interpolation between instrumented neighbours than to transfer to an uninstrumented region.

The ranking of models is unchanged by the harder partition, and the margins widen. In the held-region arm the global tree ensemble again attains lower station-median RMSE than ThermoRoute (0.652 against 0.676 °C at 1 day, 1.391 against 1.428 at 3 days, 1.786 against 1.860 at 7 days), with paired median differences of +0.031 [+0.023, +0.040], +0.038 [+0.029, +0.050], and +0.067 [+0.049, +0.080] °C and ThermoRoute win rates near 0.16. The global LSTM, with its station embedding disabled in this arm, gives 0.679, 1.445, and 1.876 °C. Since the spatial

partition moves the answer this much, the next question is whether the remaining skill is spatially and seasonally uniform.

4.5 Skill is regionally uniform, and interval coverage is bought with width

It is uniform, and this is the one place where the constrained model's behaviour is unremarkable in a useful way. Region-weighted skill against persistence — the mean of the 15 per-HUC2 medians — is +0.204, +0.189, and +0.253 at 1, 3, and 7 days, essentially unchanged from the pooled medians of +0.203, +0.187, and +0.251, so no single region carries the headline. Per-region medians against persistence at 1 day range from +0.131 (HUC2:10) to +0.285 (HUC2:18) and at 7 days from +0.217 (HUC2:06) to +0.290 (HUC2:09). Stratifying by drainage area into three groups of roughly 39 stations gives 1-day medians of +0.190, +0.214, and +0.231 from small to large. Against damped persistence the seven-day per-region medians compress to a range of +0.012 to +0.077 — the same message as Section 4.1, restated spatially: what varies across regions is small once the seasonal reference is doing its work.

Interval behaviour tells the complementary story about what calibration costs. Split-conformal intervals on the 249,072 development keys attain an empirical marginal coverage of 0.909 against a nominal 0.90, with a mean width of 3.87 °C and a mean interval score of 4.93; per lead, coverage is 0.905, 0.910, and 0.912 at widths of 2.01, 4.22, and 5.38 °C. On the warm-season tail slice defined by training-period q90 exceedance (30,531 keys), coverage is 0.918 at a width of 3.50 °C. The two pre-specified sensitivities show how much of that is a convention. Calibrating on the maximum nonconformity score within blocks of seven consecutive retained calibration rows raises overall coverage to 0.981 but widens the interval to 5.74 °C — coverage bought with width, not with sharpness. The idealized delayed adaptive-conformal variants track nominal coverage more closely as the step size grows (0.903, 0.900, and 0.892 for $\gamma = 0.005, 0.02, 0.05$) but produce unbounded widths in several slices, so their interval scores are not finite. We report width and interval score beside every coverage figure precisely so that adaptive calibration is not presented as free, and none of these figures is a conditional-coverage statement.

Input perturbations locate the model's dependence in the same two channels the ablations pointed to. Gaussian sensor noise at 0.25 and 0.5 training standard deviations degrades 1-day RMSE from 0.631 to 1.621 and 2.999 °C (+147% and +360%); missing-forcing blocks of 3, 7, and 14 days cost about +0.13 °C at 1 day and +0.04 °C at 7 days; air-temperature offsets of ± 2 training standard deviations cost +0.11 to +0.15 °C at 1 day; and multiplying discharge by 0.5 or 2 changes 1-day RMSE by at most +0.003 °C. On this panel the predictor is dominated by the water-temperature and air-temperature channels and is nearly insensitive to discharge. These are synthetic data-corruption probes, not climate projections, physically coherent scenarios, or deployment-safety tests.

4.6 Evaluation-period results

The following slots are rendered deterministically from the verified acquisition receipt after the single opening described in Section 3.7. Each is reported exactly once, including adverse, conflicting, and non-reportable outcomes, and each carries the fixed-cohort descriptive verdict of Section 3.6. Handwritten substitution into these slots is rejected by body hashes and evidence bindings.

Table 4.1 — the five formal rows. Fields: model pair, lead, predeclared margin, reportable station count, reportable cluster count, median paired station RMSE difference, whole-HUC2 cluster-bootstrap 95% interval, win rate, exact sign-flip raw p-value, Holm-adjusted p-value, and the bound gate verdict.

#	Comparison	Lead	Margin	Effect and interval	Sensitivities (p _{raw} / p _{Holm})	Verdict
1	ThermoRoute vs. damped persistence	1 d	0.00 °C	[TO BE FILLED AFTER OPENING]	[TO BE FILLED AFTER OPENING]	DESCRIPTIVE_ ONLY_ INFERENCE_ GATE_FAILED
2	ThermoRoute vs. damped persistence	3 d	0.00 °C	[TO BE FILLED AFTER OPENING]	[TO BE FILLED AFTER OPENING]	DESCRIPTIVE_ ONLY_ INFERENCE_ GATE_FAILED
3	ThermoRoute vs. damped persistence	7 d	0.00 °C	[TO BE FILLED AFTER OPENING]	[TO BE FILLED AFTER OPENING]	DESCRIPTIVE_ ONLY_ INFERENCE_ GATE_FAILED
4	ThermoRoute vs. LightGBM	3 d	+0.05 °C	[TO BE FILLED AFTER OPENING]	[TO BE FILLED AFTER OPENING]	DESCRIPTIVE_ ONLY_ INFERENCE_ GATE_FAILED
5	ThermoRoute vs. LightGBM	7 d	+0.05 °C	[TO BE FILLED AFTER OPENING]	[TO BE FILLED AFTER OPENING]	DESCRIPTIVE_ ONLY_ INFERENCE_ GATE_FAILED

Table 4.2 — all-model scores on the evaluation common keys. [TO BE FILLED AFTER OPENING] — station-median RMSE for every primary model at every lead on the exact common key registry, with the reportable station count. The frozen model registry for this cohort is the six primary models together with the seven one-factor architecture controls of Section 3.2, so the control rows are rendered here on the same keys and the same denominators; they remain exploratory deletion and intervention sensitivities and cannot promote or replace a formal row.

Table 4.3 — temporal coverage and calendar balance. [TO BE FILLED AFTER OPENING] — for each of the five rows, all eight predeclared descriptive candidates (equal-weighted 12 year-by-season cells; leave-one-year for 2021, 2022, 2023; leave-one-season for DJF, MAM, JJA, SON) and the deterministic most-adverse value.

Table 4.4 — interval and probability behaviour on the evaluation keys. [TO BE FILLED AFTER OPENING] — by model and lead: empirical marginal coverage at the nominal 90% level, mean interval width, the equal-weight three-quantile pinball mean, Brier score and Brier skill against the frozen seasonal reference, log loss, AUROC, AUPRC, expected calibration error, calibration slope and intercept, and station-balanced reliability bins. This table is filled from the evaluation-period probabilistic family computed by the trusted scorer during the single opening; it is labelled exploratory, consistent with Section 3.6, and it is a different object from the development-period suite of Section 6.3. Three fields are not produced by the opening and are rendered as explicit status tokens rather than as numbers or blanks: the interval score, the block-maximum calibration sensitivity, and the delayed adaptive-conformal sensitivity.

Table 4.5 — outcome quality control. [TO BE FILLED AFTER OPENING] — immutable counts by cohort, station, variable, raw qualifier string, and value presence; multiple-finite-series conflict counts; the outcome quality-control gate status; and the qualifier-restricted $\{A\}$ sensitivity for each formal row.

Table 4.6 — external arm on the evaluation keys. [TO BE FILLED AFTER OPENING] — the 30-site site-identifier-disjoint arm, labelled exploratory and history-dependent. The leave-HUC2-region-out arm has no evaluation-period counterpart: the opening produces no held-region artifact, so that fragment renders as an explicit status token and the whole-region holdout evidence in this paper remains the development-period evidence of Section 4.4.

5 Discussion

5.1 Why these numbers are smaller than published gains

Our reported gains are far smaller than those commonly published for daily river temperature, and the difference is a difference of design rather than of model quality. Three specific choices account for it, and each can be read off Section 4 directly.

The first is the reference. Where a study quotes skill against persistence or climatology, the number it reports is comparable to our +0.203 to +0.251 column, not to our +0.038 to +0.168 column. On a variable with this much day-to-day memory the two differ by up to a factor of 6.6 for the same fitted weights, and the discrepancy grows with lead, so studies at longer leads are the ones most affected. We would expect much of the between-study dispersion noted by Corona and Hogue (2025) to be absorbed by re-scoring published models against a fitted damped anchor rather than against persistence, and we suggest damped persistence as the minimum reference for this variable.

The second is the spatial partition. A model evaluated on randomly held-out gauges in a dense network is being asked to interpolate between instrumented neighbours; the same model evaluated on whole held-out regions, 289 km from the nearest training gauge, retains 62% rather than 92% of its three-day skill. Where published transfer results use random site splits, they are not measuring the quantity their framing implies, and the difference is roughly a third of the reported skill on this panel. This is a scale effect rather than a modelling disagreement: at the spacing of a national monitoring network, random site holdout and regional holdout are different experiments.

The third is the key set and the preprocessing boundary. Scoring each model on its own complete-case set and fitting scaling, imputation, climatology, event thresholds, or interval offsets on windows that include the evaluated interval both move error downward without leaving a visible trace. We cannot quantify how much of the published literature is affected, and we do not assert that it is; what we can say is that removing both channels here left the constrained deep model behind a gradient-boosted tree at every lead.

We are more cautious about the mechanistic reading of Section 4.3. That an information-matched plain causal convolutional network reproduces the full architecture to within $0.023\text{ }^{\circ}\text{C}$, and that deleting the router, the mixture, the dynamic prior, or the residual bound moves 1-day error by at most $0.004\text{ }^{\circ}\text{C}$, suggests that the accuracy available to a point-scale statistical predictor on this panel is largely exhausted by a causal sequence encoder over issue-time history plus a seasonal anchor. It does not identify why. Capacity matching is not search-budget matching, deletion is not attribution, and the controls are exploratory throughout.

5.2 What a benchmark of this geometry can carry

The evaluation design of this study is stronger than its inferential reach, and the two should not be confused. Clustered inference over a gauge panel rests on resampling or sign-flipping whole spatial units, and the accuracy of those procedures depends on having enough units, of comparable size, that the asymptotics they invoke are not a fiction. Fifteen HUC2 groups with an inverse-Herfindahl effective count of 9.54 and a largest group holding 21.7% of stations is not enough — and, as Section 3.6 sets out, the cluster count is the least binding of three independent reasons why this study is descriptive. The sampling design was never exchangeable across regions, the null-simulation calibration was never built, and the threshold that has attracted the most attention was written at a value no U.S. cohort can reach.

The usual practice in applied water-temperature machine learning is to report a station-level or row-level confidence interval, or a paired test across sites, without addressing whether the number of independent spatial units can carry the procedure. Under that practice a cohort like ours would produce intervals and p-values that look authoritative, and nothing in the paper would tell a reader that the effective number of clusters is under ten. The difference between that paper and this one is not the data and not the model; it is whether the limit is disclosed.

The design lesson is transferable. If a study intends region-clustered inference over U.S. gauges, the cluster structure must be part of the *sampling design*, not a post-hoc grouping of whatever cohort availability produced, and the partition and its power must be fixed before any outcome is accessible. Reaching a genuinely large

number of independent spatial units requires either a finer partition with a defensible independence argument or a deliberately stratified draw across many more regions; neither is achievable by relabelling an existing cohort. And a threshold stated in units of a fixed national partition should be checked against how many such units exist before it is sealed. A second instance of the same discipline appears in Section 6.3, where a pre-registered data contract rejected an output we would have preferred to report and we left the contract alone. A pre-registration that never binds against its authors' convenience has not been tested; these two cases are the evidence that this one has.

5.3 What transfers

This study is narrow by construction. It is a common-key, clustered, pre-registered benchmark of point predictors at gauged sites, and it is not an attempt to replace process-based thermal models, river-network graph models, or differentiable hybrid formulations (Jia et al., 2021; Rahmani et al., 2023; Zwart et al., 2023), nor architectures that encode geographic context across regions and scales (Luo et al., 2025), each of which represents structure — connectivity, upstream forcing, energy balance, spatial context — that a point-scale statistical predictor does not. Every arm here, including the whole-region and site-identifier-disjoint arms, consumes the target site's observed water temperature through the issue date (Section 3.4), so the paper is silent on prediction at locations with no thermal record.

What is portable is not the architecture but three controls that do not depend on it: scoring every model on an identical key registry so that no model benefits from selective prediction; fitting every preprocessing and calibration statistic strictly backwards in time; and partitioning space by whole hydrologic regions rather than by site. None requires unusual computational resources, and each moved a reported number in this study by more than the difference between the competing architectures. On this evidence, benchmark design is a larger source of variation in reported river-temperature skill than model design is.

6 Limitations

6.1 Machine-verifiable scope statements

The following statements are part of the evidence contract and remain in the manuscript regardless of the eventual numerical outcome.

Route A is history-dependent and uses target-site water-temperature observations through each issue date; it does not establish ungauged prediction.

Route A does not establish independent river-network separation, physical river-network routing, or hydraulic travel time.

Route A is a one-shot retrospective historical-information evaluation, not an operational replay with archived as-issued predictor vintages or future NWP.

Route A uses daily-mean statistical thresholds and a numerical non-inferiority margin; neither has ecological, biological, or regulatory meaning.

Route A provides no physical, deployment, regulatory, or distribution-free safety guarantee, and failure to reject is not equivalence.

Route-A architecture controls and predictor sensitivities are descriptive or exploratory and do not identify causal mechanisms.

Route A evaluates a fixed availability-enriched 120-site cohort and is not nationally representative of all U.S. rivers or all calendar days.

Route A does not establish conditional coverage, and its equal-weight three-quantile pinball summary is not CRPS.

6.2 Cohort, measurement, and design limitations

The station sample is availability-enriched rather than randomly drawn, and the coverage thresholds that produced it used the 2019–2020 interval, so that interval is not independent of cohort construction. The original provider bytes for the 2006–2020 panel are unavailable, so development reproduction begins from the committed derived artifact and the 1,465-candidate discovery execution is auditable but not replayable; the panel retains no NWIS qualifier columns or method and sensor history for that period, so measurement discontinuities cannot be reconstructed

for the development record. Meteorology is represented at the station coordinate rather than integrated over the contributing catchment, which is least appropriate for the largest basins. Two stations contain 2,059 signed negative discharge rows whose semantics are unresolved. Outcomes are daily means, and the evaluation is conditioned on outcome observability. HUC2 is a coarse administrative grouping and is not an independent river-network component. The $+0.05$ °C margin is a numerical ceiling with no stakeholder-derived importance. Covariates are retrospectively acquired latest-provider values rather than as-issued vintages, and model-selection budgets were documented but not equalized across model classes. Training wall-clock time and inference throughput were not recorded for the frozen suite, and no latency, memory, energy, or multi-hardware benchmark is reported. The evidence chronology is owner-controlled, with no external timestamp or independent custodian. The air2stream-style hybrid reference was never fitted, so the hybrid process family is absent from every comparison in this paper.

6.3 The development-period probabilistic suite is not reported

This section concerns the development-period probabilistic evaluation only. The evaluation-period probability family — coverage, width, three-quantile pinball, Brier score and skill, log loss, discrimination, calibration slope and intercept, and station-balanced reliability bins — is computed by a separate code path inside the single opening and is reported in Table 4.4. Nothing below extends to it, and the two must not be read as one withheld result.

The pre-registered development-period probabilistic contract treats a zero-width nominal prediction interval as a fatal condition and prohibits evaluation-time repair. In the development panel, 135 of 26,993,675 member-level rows carrying complete quantile heads — 0.0005%, or about one row in 200,000 — have identical 5th, 50th, and 95th percentile predictions. All 135 come from the tree ensembles (123 from the global model, 12 from the per-station variant); no other model is affected. We emphasise what the artifact is *not*: there is no strict quantile-ordering violation anywhere in the panel, and the maximum monotonicity violation is exactly 0.000 °C. The rejected rows are correctly ordered but degenerate.

Their distribution is interpretable, and we flag the reading that follows as interpretation rather than as a measured attribution. The 135 rows fall on 12 distinct sites, all in hydrologic regions 04, 05, and 06 — the Great Lakes, upper Mississippi, and Missouri basins — and 129 occur at the 1-day lead, 6 at 3 days, and none at 7 days; in 95.6% of them the observed target is below 1 °C. This is the signature of ice-affected winter conditions, in which daily mean water temperature is pinned near the freezing point and the conditional percentiles genuinely coincide. The 12 per-station cases are constant-leaf predictions from a model fitted on a small station-specific sample. The model outputs are defensible; it is the contract's inclusive equality test that refuses them.

We did not amend the contract. It was frozen before any access to the evaluation labels, and relaxing it would have invalidated the frozen training receipts on which the chronology gate of Section 3.7 depends. The development-period probabilistic metric suite — interval coverage from the nominal heads, three-quantile pinball mean, reliability, and Brier score over 2019–2020 — is therefore not reported, and the analysis stage that consumes it is likewise not reported. Two things follow. Development-period interval validity rests entirely on the split-conformal analysis of Sections 3.5 and 4.5, which is unaffected: the conformal offset is non-negative by construction and the deployed contract separately requires positive width, so no delivered interval is degenerate. And the standard event-probability verification metrics (Wilks, 2011) are unavailable for the development period, so no development-period claim about probabilistic calibration beyond empirical marginal interval coverage should be read into this paper; the corresponding evaluation-period metrics are reported in Table 4.4. Recovering the development-period suite requires a sealed erratum and a new training lineage.

6.4 Threshold and archive scope

An optional threshold analysis is separate from prediction evaluation. It can only compare independently sourced observed daily maxima, expressed as a strict seven-consecutive-day average of daily maxima, against a complete site-specific standards registry, with seasonal applicability assigned by each window's ending date; season-external or incomplete windows remain unclassified, and ambiguous standard

matches fail closed. Any reported exceedance is descriptive: it is an observation, not a model result and not a legal determination.

The current archive profile is machine-marked local-owner evidence: public redistribution and third-party transfer are disabled and the public build mode fails closed pending a byte-bound rights review, so this profile must not be read as a public software or data release. The self-contained history bundle deliberately retains reachable deleted objects so that chronology can be audited; it is not a byte-level purge, and those objects are provenance rather than current evidence.

7 Conclusions

We evaluated daily river water-temperature prediction at 120 stable U.S. gauges across 34 states and 15 hydrologic regions over 2006–2020, scoring every model on one common set of 249,072 station/date/horizon keys at 1-, 3-, and 7-day leads, with every preprocessing and calibration statistic fitted strictly backwards in time, and with the spatial partition varied from seen stations to random held-site folds to whole held-out regions.

Three quantitative answers follow. Reported skill depends far more on the reference model than on the architecture: median station skill at a seven-day lead is +0.251 against persistence and +0.038 against damped persistence, and of the 0.560 °C reduction in station-median RMSE from persistence to the deep model, 0.479 °C is delivered by seasonal damping alone. Model ranking does not favour architectural constraint: a gradient-boosted tree with site identity has the lowest station-median RMSE at all three leads and at every station at a one-day lead, and an information-matched plain causal convolutional network reproduces the full architecture to within 0.023 °C, while deleting the router, the mixture, the dynamic prior, or the residual bound moves one-day error by at most 0.004 °C. Reported spatial transfer depends on how the map is cut: three-day skill against persistence falls from +0.187 to +0.172 under a random held-site split and to +0.116 under whole-region holdout, at a mean nearest-training-gauge distance of 289 km, on a panel where 19 of 120 stations have a neighbour within 10 km.

The mechanism common to all three is that each design choice absorbs an alternative explanation into the reported number: seasonal damping into a weak reference, selective prediction into a model-specific key set, and spatial interpolation into a random site split. For water-resource practice this matters because reported skill is what a manager compares when choosing a tool. On this evidence, a published river-temperature skill score is not interpretable without its reference model and its spatial partition, and the incremental value of architectural complexity over a strong statistical anchor and a well-tuned tree is small enough that it should be re-measured under these controls before it is relied upon.

The formal comparisons of this study are permanently descriptive. At most 15 HUC2 groups with an inverse-Herfindahl effective cluster count of about 9.54 cannot pass the outcome-free inference gate, and two further components of that gate — the structural sampling assumptions and an unimplemented null simulation — fail independently of cohort size, so the five comparisons are fixed-cohort descriptive effects under `DESCRIPTIVE_ONLY_INFERENCE_GATE_FAILED`, with whole-HUC2 bootstrap intervals, exact sign-flip p-values, and Holm adjustments retained only as assumption-conditional sensitivities. Exploratory and secondary analyses remain reportable as such and are not erased by the gate. No result here is a superiority, non-inferiority, equivalence, or parity statement, and none generalises to a national or U.S.-river population.

Open Research Statement

Data availability. The derived daily panel (`panel.usgs.120v2.parquet`), the stable station registry (`station_registry_v1.csv`), the hydrologic-unit metadata snapshot, the candidate-rejection ledger, and the frozen panel manifest that binds them are deposited as one versioned dataset at [DATA DOI TO BE MINTED] under [DATA LICENCE TO BE ASSIGNED]. The evaluation-period acquisition record — exact request and response bytes, series identifiers, approval qualifiers, final URLs, retrieval timestamps, and content hashes — is added to that deposit as a new version after the acquisition described in Section 3.7.

Neither the DOI nor the data licence can be assigned before the byte-level rights review described in Section 6.4 completes. The derived panel encodes values from three providers whose terms differ, and a derived product does not inherit the most permissive of its upstream terms. Until every object proposed for the deposit carries a recorded, evidence-backed redistribution decision, the project's release tooling refuses to build a public archive. The deposit is therefore described here as planned and specified, not as existing.

Primary observational sources. All observations are obtained from public providers and none is the property of the authors. Daily-value water temperature and discharge come from the U.S. Geological Survey National Water Information System (U.S. Geological Survey, 2024; parameter codes 00010, 00060, and 00065 with statistic code 00003). Air temperature, precipitation, the relative-humidity proxy, and daylight-period mean incoming shortwave radiation come from Daymet V4 R1 at ORNL DAAC (Thornton et al., 2022). Wind speed comes from gridMET (Abatzoglou, 2013). Each retains its provider's own terms, citation requirement, and access route independently of this manuscript.

Material that cannot currently be redistributed. For any provider object whose redistribution terms are unresolved at deposit time, the deposit carries, in place of the bytes: the product identifier and version, the exact request specification, the retrieval code, and a SHA-256 manifest of the bytes as retrieved, so a third party can re-acquire identical inputs and verify them without relying on redistribution. The classes currently defaulting to exclusion, and the two excluded outright — the third-party AGU LaTeX class, and three legacy CSV files retained in the repository archive from an earlier three-station case study whose source and collection terms are unrecorded — are enumerated in SI16; the legacy material is not evidence for any statement in this manuscript and its status is recorded in `protocols/legacy_three_site_semantics_notice_v1.md`. Original provider responses for the 2006–2020 development panel were not retained, so development reproduction begins from the committed derived artifact; that limitation does not apply to the evaluation-period acquisition, for which exact bytes are archived.

Software availability. The analysis software, the frozen protocol and its sealed amendments, the probability-metric erratum, the model-suite registry, the

completion receipts, the per-stage manifests, the environment probe, the verification entrypoints, the deterministic result renderer that generates Section 4.6 from the acquisition receipt, and the fully transitive Python 3.12 dependency lock with package hashes are archived at [SOFTWARE DOI TO BE MINTED], corresponding to release [RELEASE TAG TO BE ASSIGNED] of the source repository at [REPOSITORY URL TO BE CONFIRMED]. The project's own source code is released under the MIT licence (SPDX identifier MIT). That licence covers the source code only: it does not license the observational data, the archived provider responses, the third-party typesetting class, or any redistributed dependency binary, each of which retains its own terms and is enumerated with those terms in an archive notice file.

Analyses were run on Python 3.12 with NumPy, pandas, pyarrow, SciPy (Virtanen et al., 2020), scikit-learn (Pedregosa et al., 2011), LightGBM (Ke et al., 2017), PyTorch (Paszke et al., 2019), and the USGS `dataretrieval` client (Hodson and Hariharan, 2023).

Reproduction. Sections 4.1–4.5 are reproducible from the archived panel, the archived model bundles, and the pinned environment. Section 4.6 is reproducible from the archived acquisition record and the same bundles. Bit-level equality is expected for the tree models and agreement to a documented tolerance for the neural members; the tolerances, the verification commands, and the expected digests are listed in the Supporting Information. Reproduction has not yet been carried out by an operator independent of the authors on an independent host, and this statement is not a claim that it has.

Placeholder inventory. Seven bracketed identifiers appear above: [DATA DOI TO BE MINTED], [DATA LICENCE TO BE ASSIGNED], [SOFTWARE DOI TO BE MINTED], [RELEASE TAG TO BE ASSIGNED], [REPOSITORY URL TO BE CONFIRMED], [TRAINING WALL-CLOCK TO BE RECORDED], and [INFERENCE COST TO BE RECORDED]. None is a real identifier or value, and none may be replaced by a reserved, draft, example, or estimated figure. Together with the author block, the corresponding-author details, and the Acknowledgments below, they are the complete set of fields that must be closed before submission; each is tracked, with the input required to close it, in `docs/WRR_SUBMISSION_CHECKLIST.md`.

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[COMPUTATIONAL RESOURCES TO BE COMPLETED — the facility or facilities on which the model suite was fitted.]

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Competing interests. [COMPETING INTERESTS TO BE COMPLETED — a declaration is required from every author, including the explicit statement that none exists if that is the case.]

Author contributions. [CREDIT ROLES TO BE COMPLETED — assign each author to the applicable CRediT roles: conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing — original draft, writing — review and editing, visualization, supervision, project administration, funding acquisition. The signed intake form is `docs/FAIR_SUBMISSION_READINESS_AND_TEMPLATES.md` §2.]

Supporting Information

Supporting Information accompanies this manuscript and contains: the cohort and registry description with the candidate-rejection ledger (SI01); the issue-time information boundary, the product-compatibility bridge, and the isolation, publication, durability, and adversary-model details of the replay and acquisition machinery referred to in Sections 3.3 and 3.7 (SI02); model equations and unit conventions (SI03); the sealed protocol, its amendments, and the probability-metric erratum (SI04); the registered five-row comparison set and its fields (SI05–SI06); all-model scores on the exact common keys, including the status, provenance, and unofficial character of the unfitted air2stream-style hybrid reference (SI07); probability and reliability schemas with the non-reporting disposition of Section 6.3 and the full measured degenerate-interval table (SI08); architecture and information-

matched controls with seed and budget provenance (SI09); the temporal coverage audit (SI10); spatial and leave-cluster sensitivities, including the recomputed cluster geometry at HUC2, HUC4, HUC6, and HUC8 (SI11); outcome quality control and qualifier evidence (SI12); the history-dependent external arm (SI13); missingness and failure cases (SI14); reproduction hashes, commands, and environment parity fields (SI15); and the rights and data dictionary (SI16).

Ten supporting figures accompany these sections: Figures S1–S3 describe the frozen cohort geometry, the temporal roles and issue-time information boundary, and the model and calibration dataflow, and contain no evaluation-period quantity; Figures S4–S8 and S10 expand Section 4.6 and are rendered only after the acquisition of Section 3.7; and Figure S9 is an optional development-period conformal-calibration sensitivity, labelled as such in the figure itself, which must not be compared numerically with any evaluation-period figure.

Each figure carries evidence from exactly one period. Figures 1 and S1–S3 are pre-opening structural material, Figures 2 and 4 and Figures S4–S8 and S10 are evaluation-period, and Figure 3 and Figure S9 are development-period and say so inside the figure. No value from one period is compared numerically with a value from another.

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